

Acquisition and Analysis of Neural Data

Bernstein Center for Computational Neuroscience Berlin

Lecturer: Prof. Dr. Richard Kempster

Tutors: Stefano Masserini, Ikhwan Bin Khalid



Problem Set 2

Deadline: Tuesday 28th April, 2026 at 23:55

Please work in groups of ideally 2 students. Every group member must submit a copy of their group solutions and include the names of all group members in the files. Please specify how much time you needed to finish each part and a statement on whether you used AI tools (otherwise, 5 points will be subtracted).

Analytical Part: scan and upload your solution as a single PDF. Make sure all text is clearly readable.

Numerical Part: submit both a Python code file (or notebook) and a PDF or HTML file including figures and code. Use clear, descriptive variable names and comments.

Analytical Part - Convolution and coefficient of variation

1. Convolution and cross-correlation (3 points)

The operations of convolution and cross-correlation are conceptually similar but have different definitions and interpretations.

The convolution between two real signals $x(t)$ and $y(t)$ is defined as

$$(x * y)(t) = \int_{-\infty}^{\infty} ds x(s)y(t-s).$$

If, for example, $x(t) = \rho(t)$ is the neural response function and $y(t) = w(t)$ is a normalized averaging window, then $(x * y)(t)$ is a time-averaged firing rate.

Similarly, the cross-correlation between two real signals $x(t)$ and $y(t)$ is defined as

$$(x \star y)(t) = \int_{-\infty}^{\infty} ds x(s)y(t+s).$$

The cross-correlation is used as a measure of similarity between two waveforms as a function of a relative time shift t between them.

- How are convolution and cross-correlation related to each other? Write the cross-correlation $(x \star y)(t)$ as a convolution $(\tilde{x} * y)(t)$. How is the new function $\tilde{x}$ related to the original function x ? When are the two operations equivalent? (Hint: use variable substitution).
- Consider the functions $x(t) := [(t/t_{\text{peak}})e^{-t/t_{\text{peak}}}]_+$ (alpha function) and $y(t) := \delta(t - t_{\delta})$ (Dirac delta). Compute the convolution $(x * y)(t)$ and the cross-correlation $(x \star y)(t)$. Sketch the result.
- Suppose you find a pronounced peak in the cross-correlation between two spike trains. How would you interpret this finding?

2. Normalization of a probability density (2 points)

A probability density function $p(x)$ is normalized such that $\int_{-\infty}^{\infty} dx p(x) = 1$. The alpha function is often used for filtering. Consider the alpha function $x(t) = [Ate^{-\omega t}]_+$. Find the scaling factor A such that $\int_{-\infty}^{\infty} dt x(t) = 1$. Hint: Use integration by parts and l'Hôpital's rule.

3. Coefficient of variation of a Poisson process (4 points)

The Poisson process is a **continuous-time counting process** where events are **statistically independent from each other**. The Poisson process is used for example **to model stochastic neuronal firing**, the arrival of customers in a queue, or the number of photons hitting a photodetector. A Poisson process with rate r has inter-event intervals τ that are exponentially distributed:

$$p_{\text{ISI}}(\tau) = \begin{cases} re^{-r\tau} & \text{for } \tau \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

- (a) Show that $p_{\text{ISI}}(\tau)$ is a normalized probability density.
- (b) Compute the mean $\langle \tau \rangle$ and the variance σ_τ^2 of the inter-event intervals τ .
- (c) Compute the coefficient of variation CV of the Poisson process. Remember $CV = \sigma_\tau / \langle \tau \rangle$.

Hint: recall that the mean $\langle x \rangle$, the second moment $\langle x^2 \rangle$, and the variance σ_x^2 of a random variable x are defined as follows:

$$\langle x \rangle := \int_{-\infty}^{\infty} dx x p(x) \quad ; \quad \langle x^2 \rangle := \int_{-\infty}^{\infty} dx x^2 p(x) \quad ; \quad \sigma^2 := \langle x^2 \rangle - \langle x \rangle^2 .$$

4. Moments of the Poisson Distribution (1 point)

Go through Appendix B of the first chapter in the Dayan and Abbott book. Make sure you understand the moment estimation of the Poisson distribution.

Numerical Part - Trial-Averaged Firing Rate and Spike-Triggered Average

1. Trial-Averaged Firing Rates (7 points)

Download `SpikeTimes.dat` from the Moodle webpage and load it with the numpy function `loadtext`. The file contains a matrix of spike times in milliseconds. The dimensions are $[m \times n]$, where $m=100$ is the maximum spike-time index and $n=100$ is the number of trials. Note: each trial has a different number of spikes, the remaining entries are filled with `nan`. The trial length is $T = 5.5$ s, the resolution is 0.1 ms. It is convenient to convert the data in units of seconds.

- (a) Make a raster plot using the pyplot function `eventplot`. You may need to set the axes limits manually for correct visualization.
- (b) Calculate and plot the trial-averaged firing rate using a sliding rectangular window with a width of 30, 90, and 200 ms. Compute the trial-averaged response function first and then convolve it with the rectangular window.
- (c) Include different numbers of trials (10, 50, 100). Comment on your results.

2. Spike-Triggered Average (3 points)

Download `STA_data.mat` from the Moodle website and load it with the `scipy.io` function `loadmat`. The file contains the stimulus vector and the spikes times of a model neuron that was exposed to the stimulus for 100 trials. The trial length is $T = 1$ s, the time resolution is 0.1 ms.

Calculate the spike-triggered average

$$C(\tau) = \left\langle \frac{1}{n} \sum_{i=1}^n s(t_i - \tau) \right\rangle = \frac{1}{\langle n \rangle} \int_0^T dt \langle \rho(t) \rangle s(t - \tau)$$

where $s(t)$ is the stimulus, $\rho(t)$ is the neural response function, and the angular brackets denote trial averages. Plot the spike-triggered average for $0 \leq \tau \leq 50$ ms. Ensure that the STA is plotted according to the convention: τ positive and decreasing from left to right. To compute the STA, you can either loop over the spike times t_i (first expression) or use the numpy function `correlate` (second expression). Both procedures shall lead to approximately the same result. Which one is more efficient?